Free pointwise control and shared parameter responses answer different questions

Free gate floors use all 160 anchors; kernels use 80 calibration anchors. Calibration output dimensions: Ring8 160; Product16 1280.

Controller P Full FFN

Ring8 · step 1000

Ring8 · step 5000

Product16 · step 1000

Product16 · step 5000

Pointwise free-gate residual energy share

Native calibration kernel mean diagonal (log scale)

Kernel participation rank (tr K)² / tr(K²)

S mul S add V mul V add Swi96 Swi128

≈0 ≈0 ≈0 ≈0 n/a n/a

0.614 0.602 ≈0 ≈0 n/a n/a

0.700 0.704 ≈0 ≈0 n/a n/a

Native coordinates only; the RMS-equalized counterfactual is excluded here. Effective rank is spectral concentration, not algebraic rank or a training guarantee.